

Energy Consumption in a Building

1. Introduction

This project analyzes energy consumption using data preprocessing techniques.

2. Objectives

Data Cleaning, Outlier Detection, Correlation Analysis

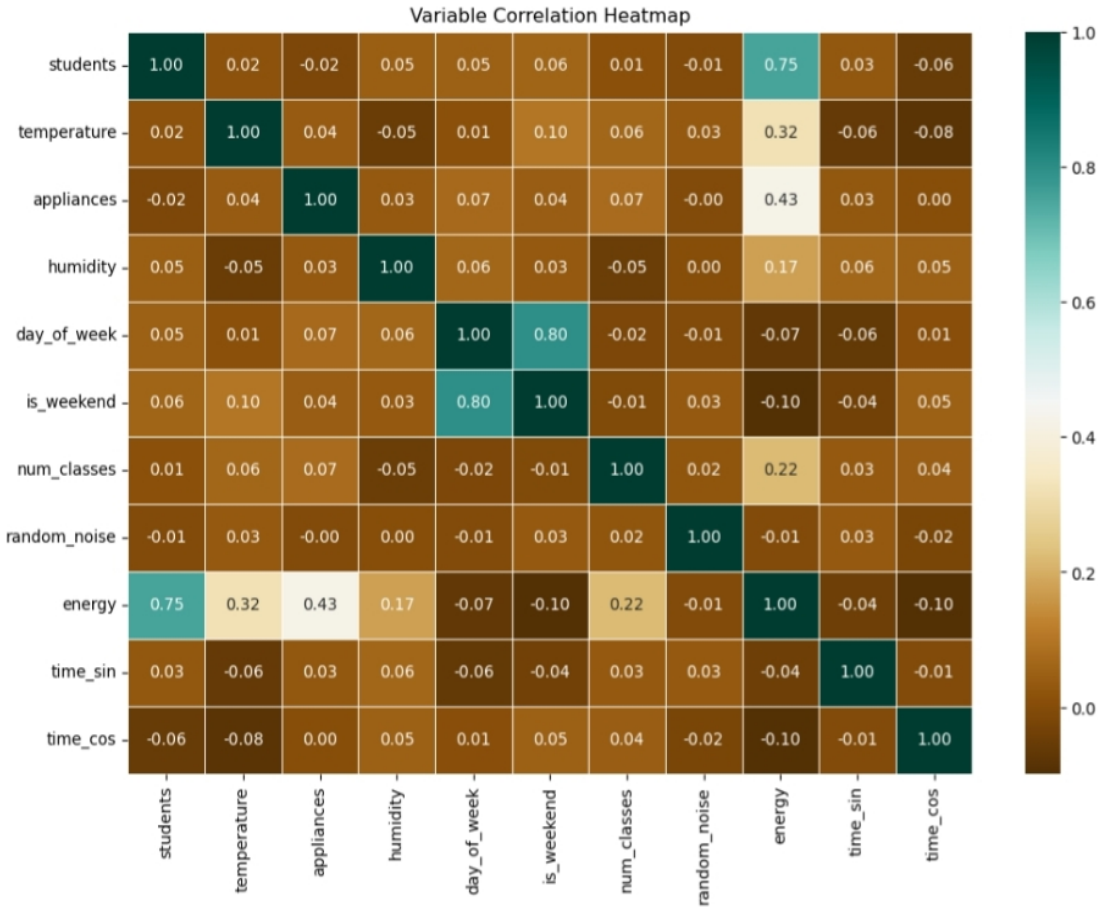
3. Methodology

3.1 Data Cleaning

Duplicate values removed and missing values handled.

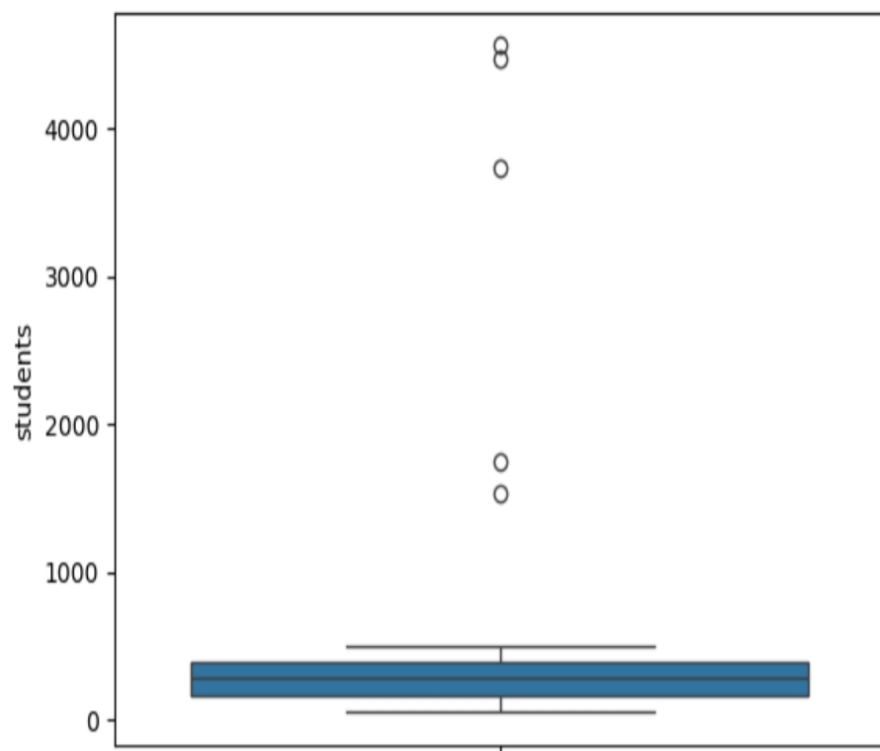
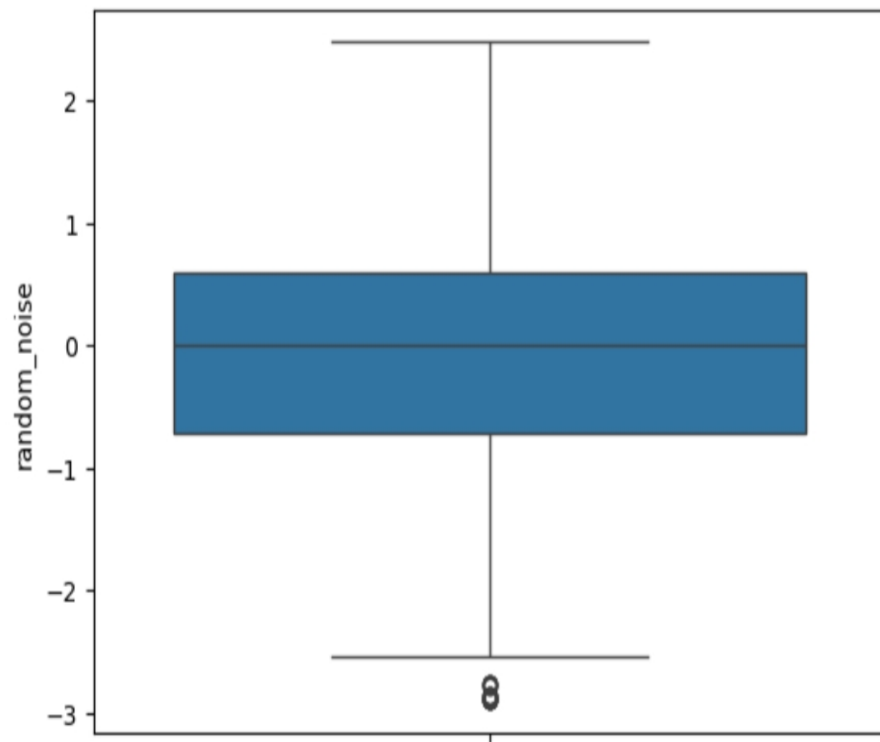
3.2 Correlation Analysis

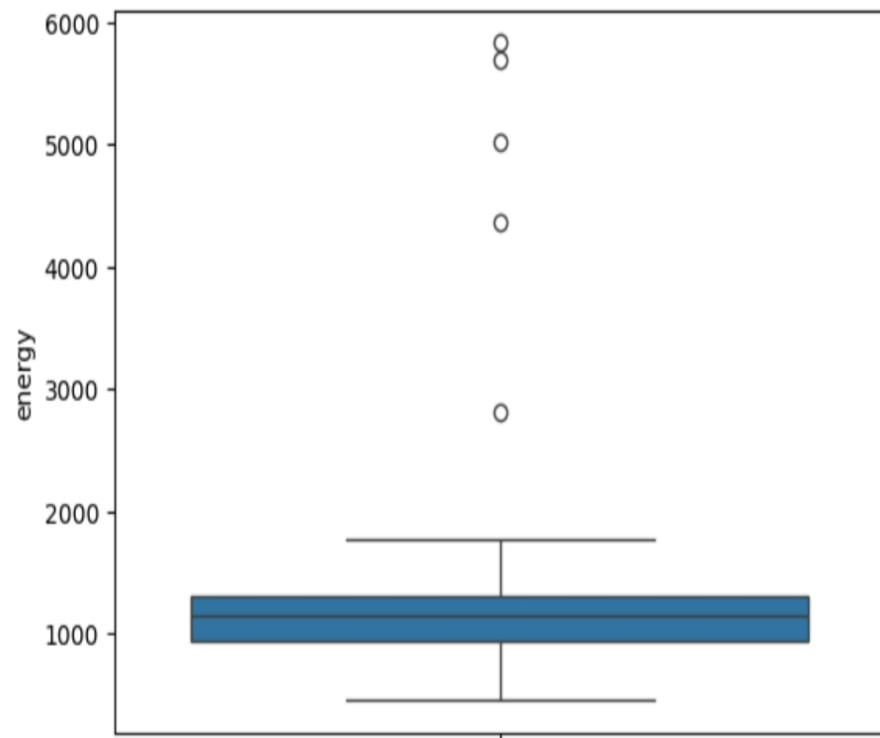
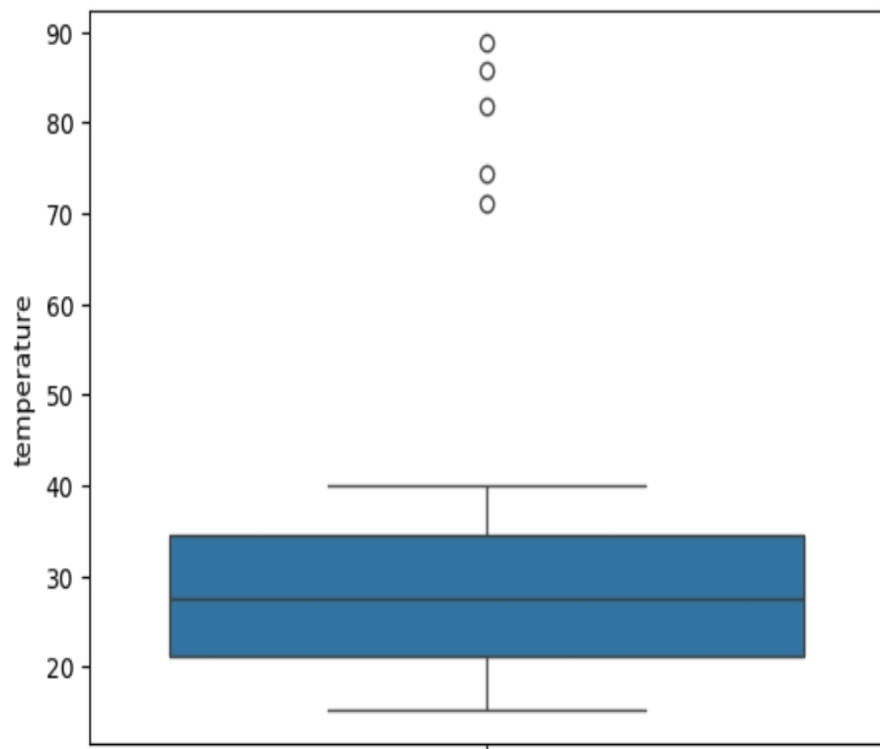
Correlation heatmap is used to identify relationships between variables.



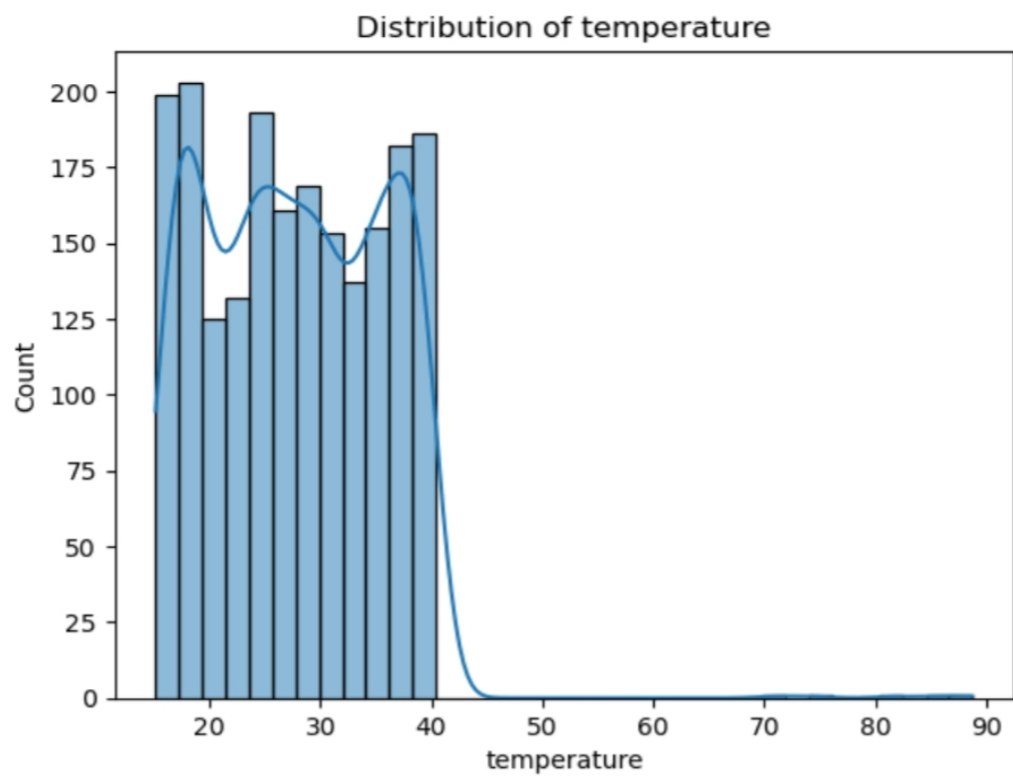
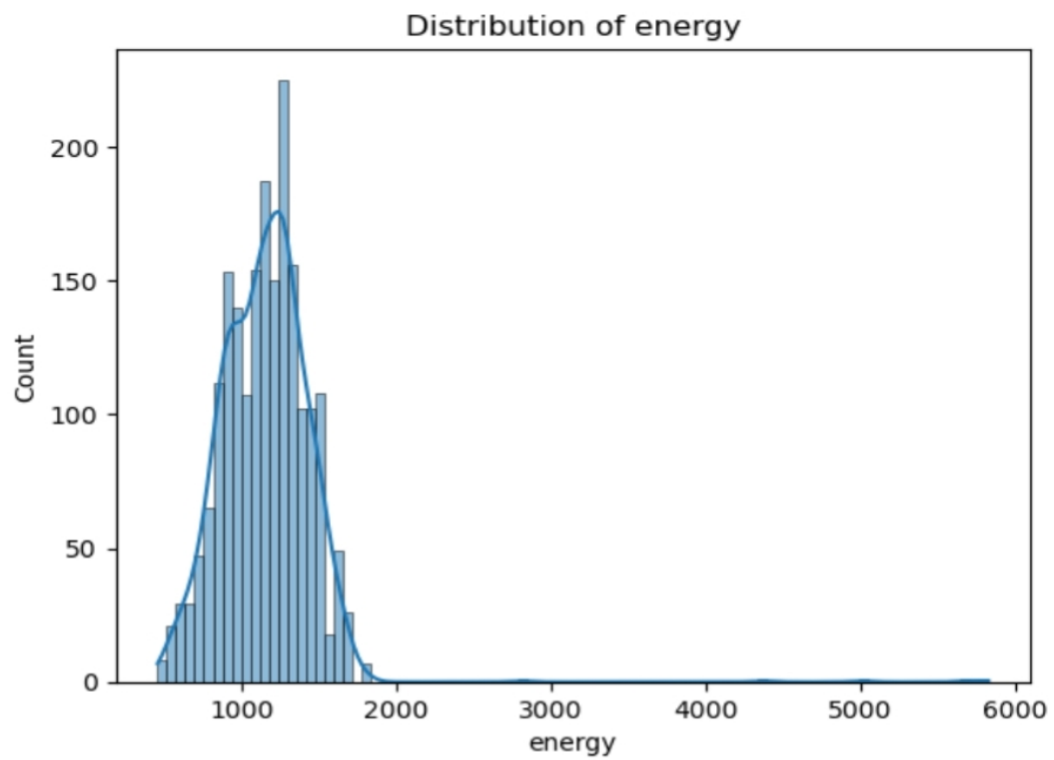
3.3 Outlier Detection

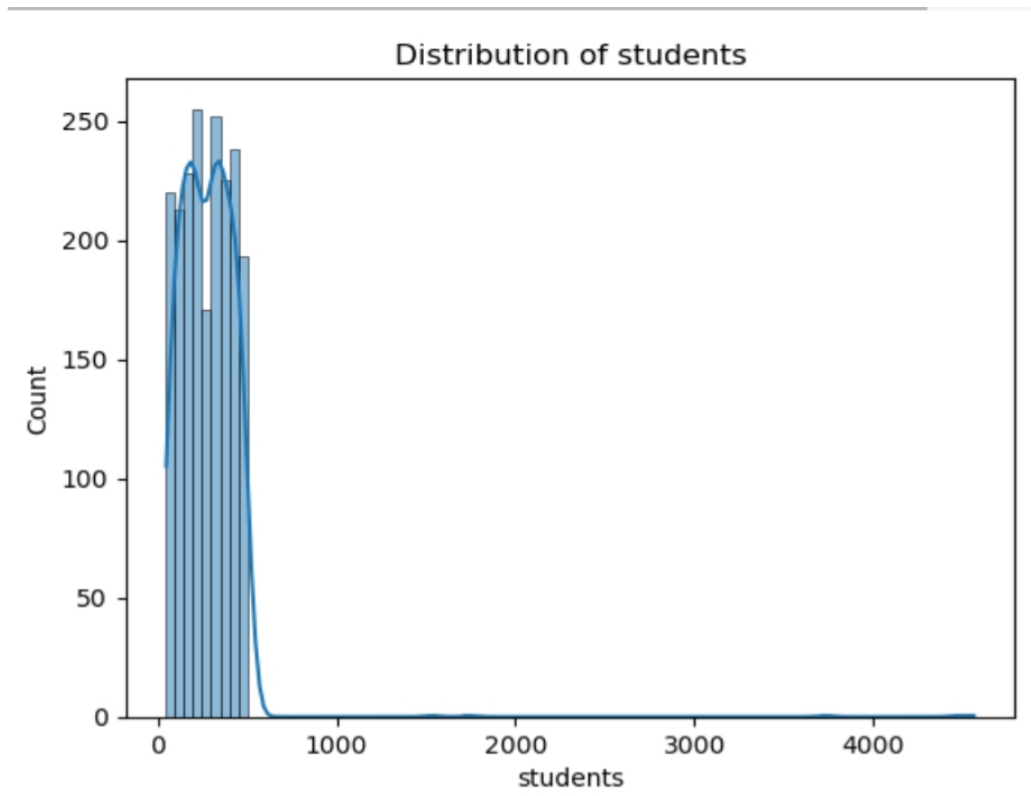
$IQR = Q3 - Q1$





4. Feature Distribution





5. Conclusion

In this project, data cleaning has been successfully performed to remove inconsistencies such as missing values and duplicate records. Outlier detection using the IQR method helped in identifying extreme values that could affect the analysis. The correlation heatmap provided clear insights into relationships between different features, especially showing how factors like students and temperature influence energy consumption. Additionally, feature distribution graphs helped in understanding the spread and behavior of the data. Overall, the preprocessing steps improved the quality and reliability of the dataset, making it suitable for further analysis and model building. This project highlights the importance of proper data preprocessing in achieving accurate and meaningful results.